

Module 5: Pragmatic & Discourse Processing

**Discourse: Reference Resolution,
Reference Phenomena, Syntactic &
Semantic constraint on coreference.**

What is pragmatics?

Pragmatics is the study of how speakers use language in context and how listeners interpret meaning beyond the literal words.

Eg., It's cold in here.

Pragmatics studies how context contributes to meaning. It includes the study of:

1. Speaker meaning
2. Contextual meaning
3. How more gets communicated than said
4. The expression of relative distance

What is discourse ?

A language does not consist of isolated , unrelated sentences but instead it consist of collocated related group of sentences . Such a group of sentences is called a discourse .

A discourse can be of the three forms:

- 1) **Monologue** (speaker → hearer or readers)
- 2) **Dialogue** (a conversation with a friend)
- 3) **Human Computer Interaction** (HCI)

Example 1:

John went to Tata Motors to check out a Volkswagen Taigun .

He looked at it for about an hour.

What do pronouns he and it denote ?

Example 2 :

I'd like to get from Mumbai to Dehradun , on either 20th or 21st of May .It's okay if it stops in another city along the way.

Terminologies used:

A natural language expression used to perform reference is called a **referring expression** , and the entity that is referred to it is called the **referent** . Here John is the referent.

Two referring expressions that are used to express to the same entity are said to **corefer** here John and he are coreferring entities.

An **antecedent** is the noun or phrase to which a pronoun or pro-form replaces it.

In the given sentence John is the antecedent of he.

Reference Resolution

It is a task of determining what a word or phrase is useful in a text.

It helps machines understand relationships between words and entities.

Example 1 : Simple pronoun resolution

Neha went to the shop. She brought chocolates.

She refers to Neha

NLP resolves she as Neha.

Example 2 : Multiple Entities

Alice gave the book to Mary because she had finished reading it.

Here she is ambiguous because it refers to either Alice or Mary.

Example 3 : Object coreference

I saw a Mercedes parked outside. It was black in color.

Here it refers to a car which is a mercedes.

Example 4: **Anaphora** Resolution

Anaphora resolution is the process of determining the referent, which is a word or phrase that refers back to something that has been mentioned earlier in the text.

The **cat** chased the **mouse** . **It** ran away.
Here it refers to the mouse. Here it resolves it as mouse.

Example 5:

Cataphora is a word or a sentence that appears latter in the sentence or discourse but its reference is made before.

Basic Structure :

Pronoun → Noun

Example

When **she** entered the room, **Nicky** was nervous.

She appears before the noun Nicky it refers to

Here she that is Nicky comes first in the sentence.